

CablePro

Cable TV Management System

A Progressive Web App for Cable TV Operators in India

Next.js 14 · React 18 · Firebase · TypeScript · Tailwind CSS v4

Abstract

CablePro is a purpose-built, web-based Cable TV Management System designed specifically for local cable TV operators in India. The cable television industry serves millions of households across the country, and with rapid government-mandated digitization, operators face increasing complexity in managing subscriber data, payment collections, and regulatory compliance.

This modern, mobile-first Progressive Web App (PWA) replaces manual registers and spreadsheets with centralized subscriber management, automatic outstanding calculations, real-time payment analytics, and secure cloud backup to Google Drive — accessible from any device, anywhere.

- **Cable Operators**

Target Users

- **PWA / Web App**

Platform

- **localStorage**

Primary Storage

- **Firebase Auth**

Authentication

Introduction

What is CablePro?

CablePro is a centralized, web-based platform enabling cable TV operators to manage all aspects of their business — from adding subscribers to tracking monthly payments and generating financial reports.

Mobile-First

Built as a Progressive Web App (PWA), it works seamlessly on smartphones, tablets, and desktops, giving operators real-time access in the field.
Designed for field use on smartphones.

Cloud Backup

Google Drive integration for data safety and secure cloud backup and recovery.

Secure Auth

Firebase Authentication with Google OAuth sign-in for secure access.

Real-Time Analytics

Dashboard KPIs and visual charts for instant business insights.

Objectives of the Project

01

Centralized Subscriber Management

Single platform for complete customer information including STB number, plan details, and subscription status tracking.

03

Payment Tracking & Analytics

Record cash and UPI transactions with real-time visual charts showing collection trends and payment methods.

05

Mobile-First Design

Flawless mobile experience optimized for field payment collection on smartphones and tablets.

02

Automated Outstanding Calculation

Auto-calculate dues per subscriber based on plan type and payment history, eliminating manual errors.

04

Financial Reporting

Generate PDF and CSV reports for daily, weekly, and monthly collections with custom date ranges.

06

Cloud Data Backup

Google Drive backup with configurable auto-frequency for data protection and disaster recovery.

Scope of the Project

1

User Authentication

Email/Password login, Google OAuth integration, profile management, and 7-day grace period for account deletion.

2

Customer Management

Full lifecycle management: add, edit, soft-delete customers with STB tracking, plan duration (1/3/6/12 months), and status filters.

3

Payment Management

Record cash and UPI transactions, track today/weekly/monthly totals, and prevent duplicate billing cycles automatically.

4

Financial Reporting

Generate reports for month or custom date ranges, analyze collection trends, and export to CSV and PDF formats.

5

Dashboard Analytics

Live KPI cards showing total customers, active subscribers, outstanding amounts, and paid collections with bar and pie charts.

6

Backup & Import/Export

Google Drive backup (manual or automatic) plus CSV import/export for bulk data migration and recovery.

Existing System — Limitations

Traditional methods used by cable operators create operational challenges and business risks:

Paper-Based Register

- Data loss risk from fire or water damage
- No STB or plan duration tracking
- Manual page-by-page search required
- No automatic outstanding calculation
- Cannot access remotely
- Error-prone manual recording

Spreadsheet (Excel/Sheets)

- Complex manual formula setup
- Poor mobile experience
- No authentication or access control
- Manual file transfer required
- Requires dedicated hardware

Proprietary Software

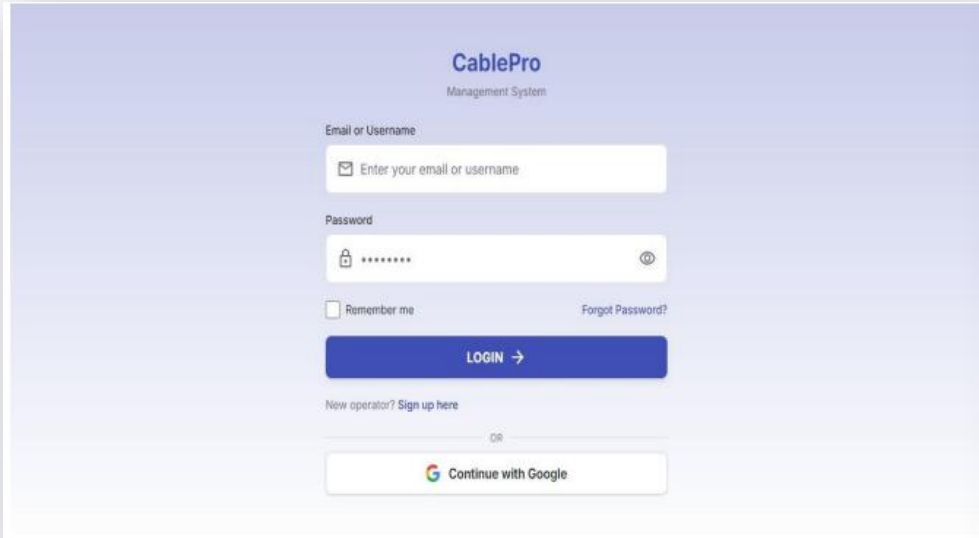
- High cost — up to ₹50,000 per year
- Not mobile-optimized
- Poor cloud integration
- Rigid system, hard to customize

Proposed System — CablePro

CablePro addresses all limitations of existing systems with modern, purpose-built features:

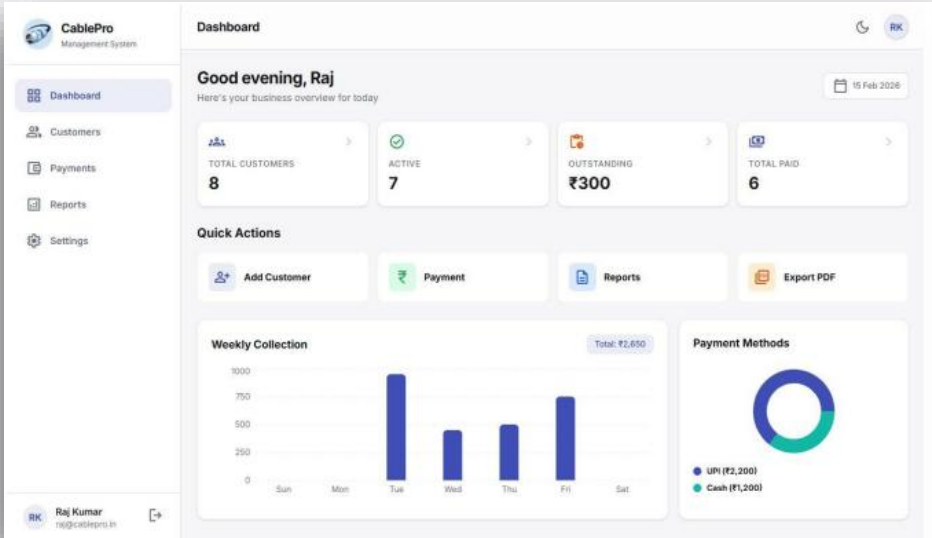
Feature	Existing System	CablePro ✓
Access	Single location/device	Any device, anywhere
Mobile Support	None / Poor	Fully mobile-optimized
Outstanding	Manual calculation	Auto-calculated
Search	Manual scanning	Instant — name/phone/STB
Reporting	Manual summarization	Automated + PDF/CSV export
Backup	None	Google Drive cloud backup
Cost	₹0 – ₹50,000/yr	Few thousand rupees per year

Screenshots — Core Screens



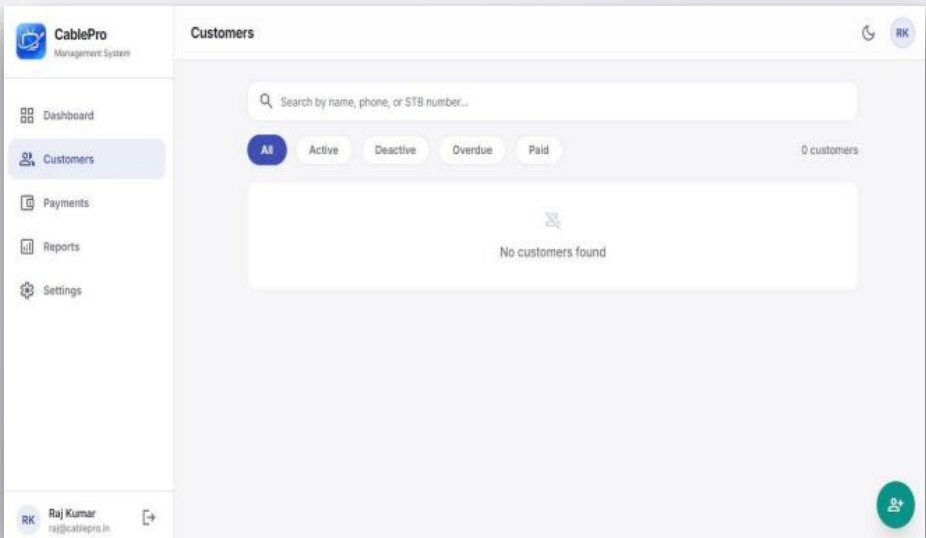
Login Page

Gradient background with email/password fields, Remember Me option, Forgot Password link, and Google Sign-In button.



Dashboard

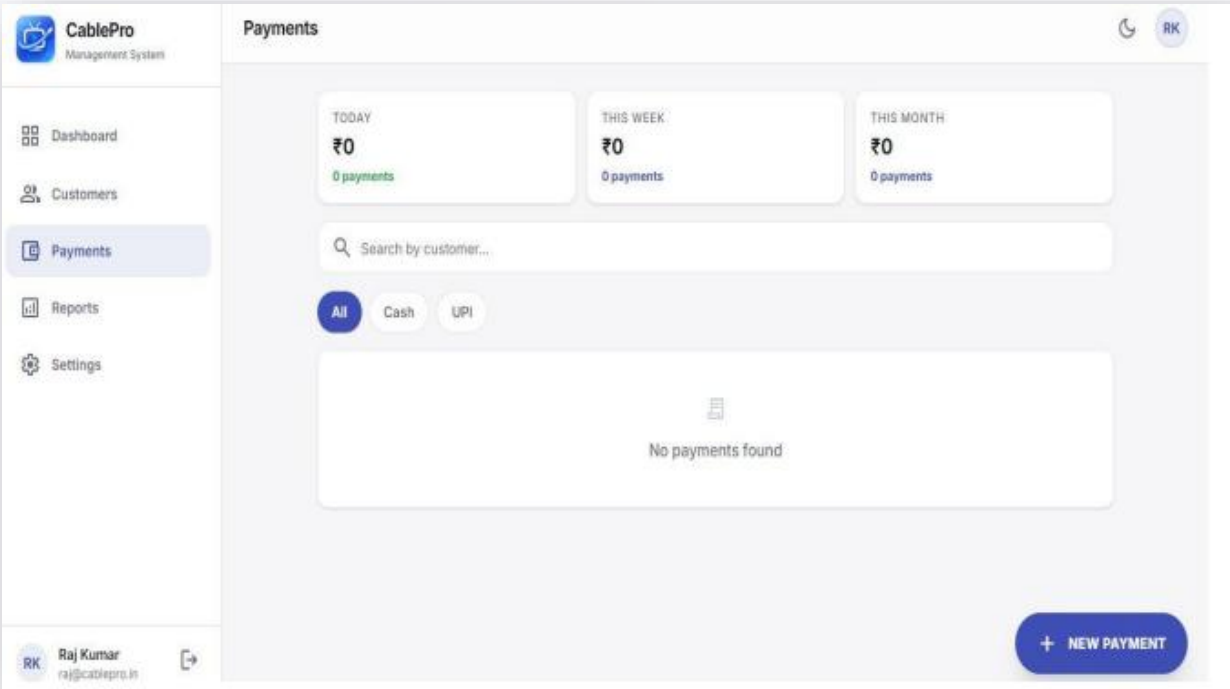
KPI cards showing Total, Active, Outstanding, and Paid customers with weekly bar chart and payment methods donut chart.



Customers Page

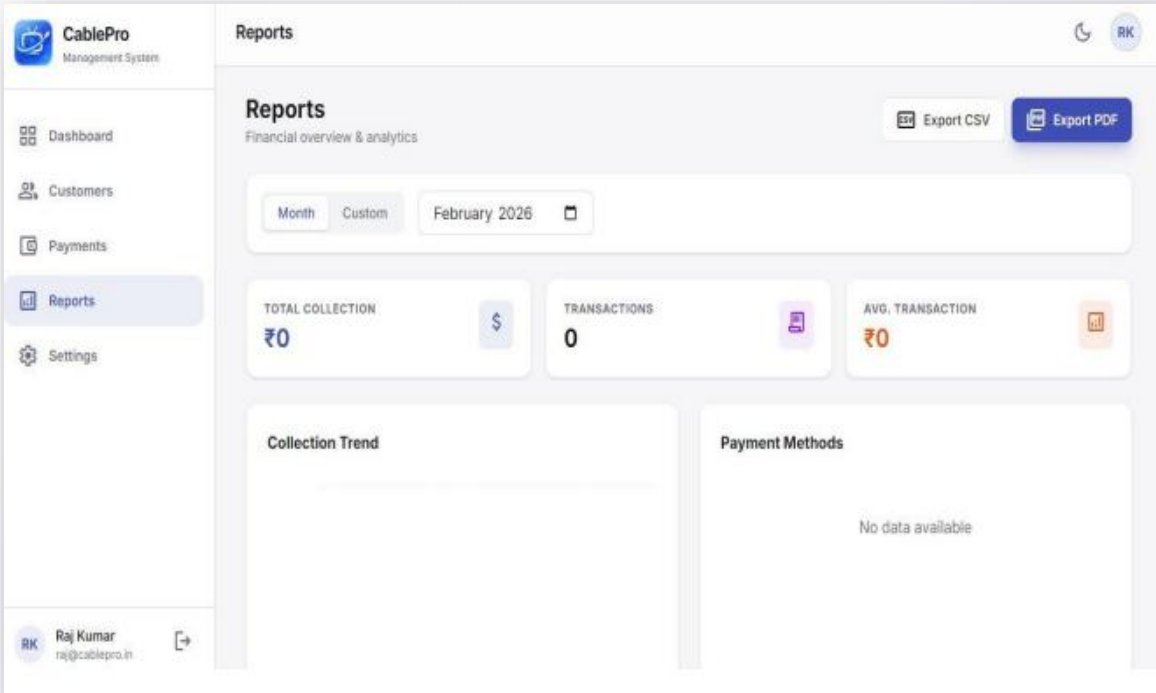
Search by name, phone, or STB number. Full customer list with status badges and action buttons.

Screenshots — Payments & Reports



Payments Page

Today/Week/Month/Total collection cards. Searchable transactions by customer or payment method (Cash/UPI), status badges, and action buttons.



Reports Page

Month or custom date range filter, KPI cards, Collection Trend chart, Payment Methods breakdown, Daily Breakdown table — exportable to CSV and PDF.

Conclusion

CablePro Successfully Addresses Real Operational Challenges



Subscriber Management

Complete lifecycle management with STB tracking and automatic outstanding balance calculation.



Smart Payments

Duplicate-proof payment module with multi-month billing cycle logic and real-time tracking.



Mobile-First PWA

Works offline with localStorage — accessible from any device, anywhere, even without internet.



Secure & Reliable

Firebase Authentication and Google Drive backup — zero hardware cost, enterprise-grade security.

Built with: Next.js 14 · React 18 · TypeScript · Firebase · Tailwind CSS v4 · Vercel